

Time-Series-Driven Predictive Maintenance

Stat 248 Final Project

May 13, 2026

Abstract

This project studies predictive maintenance as a time-series forecasting problem with an operational decision layer. A synthetic fleet of degrading machines generates vibration, temperature, and pressure signals over time. The main question is whether richer temporal summaries improve short-horizon failure-risk prediction and, in turn, lower maintenance cost. Results show that nonlinear time-series features improve predictive accuracy over raw-sensor baselines, and a simple risk-threshold policy outperforms reactive and fixed-schedule maintenance. The main lesson is that better temporal representation matters more than a more complicated controller.

1 Introduction

Predictive maintenance is a natural time-series problem because machine failure is driven by the evolution of degradation rather than by one isolated sensor reading. In this project, the main objective is to determine whether temporal summaries extracted from sensor streams improve near-term failure prediction. A secondary objective is to show that better risk estimates can be converted into better maintenance decisions.

2 Data-Generating Process

The project uses a fully synthetic but reproducible fleet simulation. Each machine has a latent health process, a stressed operating regime, and three observed sensors: vibration, temperature, and pressure. Machines fail when latent health falls below a threshold or when a high-hazard event occurs near the end of life. The supervised label at each timestamp is whether failure occurs within the next eight periods.

Table 1: Summary of the generated dataset across train, validation, and test splits.

split	n_rows	n_machines	mean_failure_time	mean_path_length	vibration_min	vibration_max	temperature_min	temperature_max	pressure_min	pressure_max
train	4726	90	51.5111	52.5111	0.2684	1.6225	57.0486	75.6074	96.5649	106.9888
validation	1723	35	48.2286	49.2286	0.2871	1.6209	58.5078	75.6872	96.9107	106.9589
test	1941	35	54.4571	55.4571	0.3233	1.5428	58.0908	75.3467	96.9284	107.4286

3 Methods

Three predictive models are compared. The baseline GLM uses current age and raw sensor values. The advanced GLM adds rolling means, rolling standard deviations, rolling slopes, and Holt level-trend summaries. The nonlinear time-series forest extends this feature set with multi-scale rolling summaries, lags, first and second differences, exponentially weighted moving averages, and stress-contrast variables.

To connect prediction with decision-making, the project compares reactive maintenance, fixed-age replacement, a tuned risk-threshold policy, tabular Q-learning, and a compact DQN extension. The RL methods are included as benchmarks, but the main focus remains on the time-series forecasting layer.

4 Predictive Results

Table 2: Predictive performance on the test fleet.

model	auc	brier	log_loss
nonlinear_ts_forest	0.8821	0.0922	0.2972
advanced_glm	0.8790	0.0935	0.2997
baseline_glm	0.8732	0.0949	0.3045

The best predictive model is `nonlinear_ts_forest`. This supports the claim that explicit temporal representation improves short-horizon failure prediction beyond current sensor levels alone.

Table 3: Top ten feature importances for the nonlinear time-series forest.

feature	importance
vibration_mean_15	0.0932
vibration_mean_8	0.0728
vibration_ewm_04	0.0627
pressure_ewm_04	0.0518
holt_level	0.0517
pressure_mean_15	0.0480
pressure_mean_8	0.0468
health_proxy	0.0412
pressure_mean_3	0.0392
vibration	0.0329

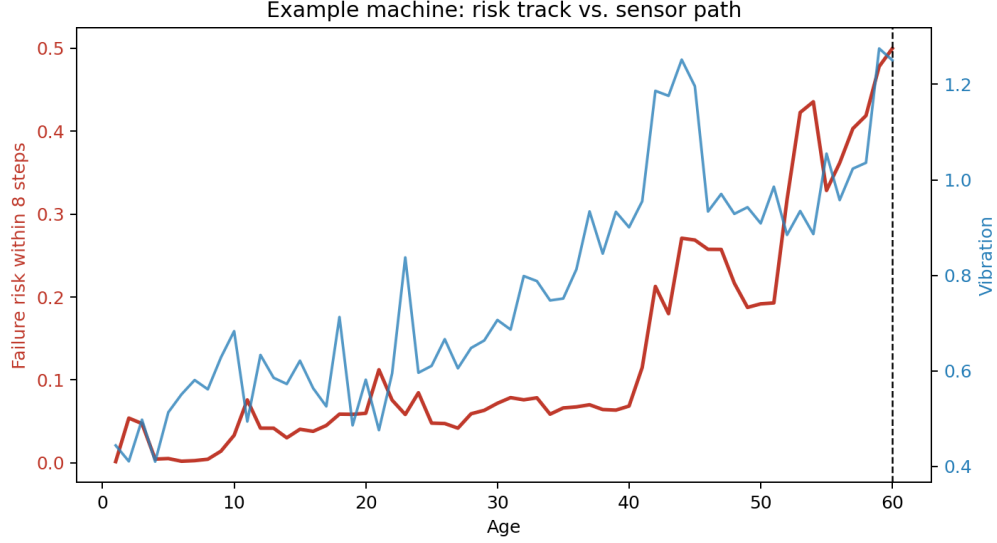


Figure 1: Predicted short-horizon failure risk for one example machine, shown alongside a representative sensor path.

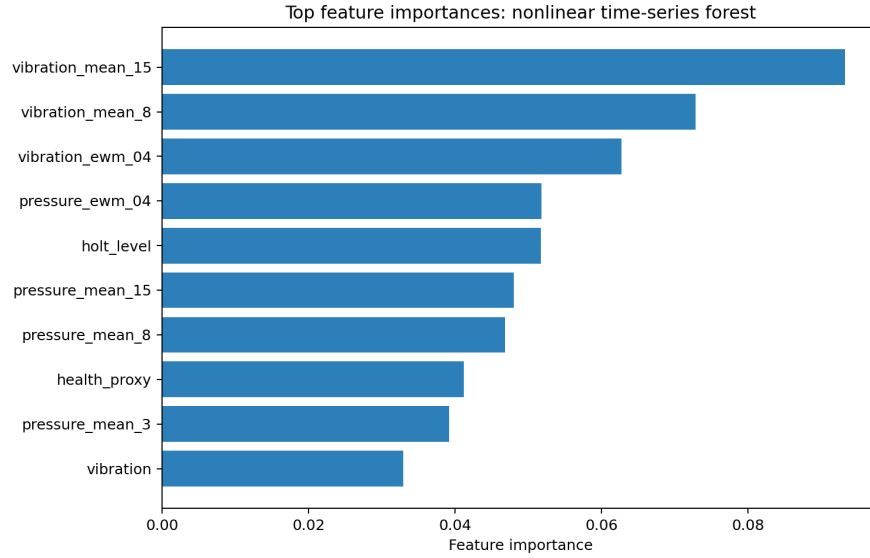


Figure 2: Top feature importances for the nonlinear time-series forest.

5 Diagnostics

To validate the time-series workflow, the project includes augmented Dickey-Fuller checks on first-differenced sensor series, calibration summaries for predicted probabilities, and residual dependence diagnostics.

Table 4: ADF diagnostics on first-differenced sensor series.

series	adf_stat	p_value	crit_5pct
vibration_first_difference	-19.0789	0.0000	-2.8622
temperature_first_difference	-18.2645	0.0000	-2.8622
pressure_first_difference	-22.7867	0.0000	-2.8622

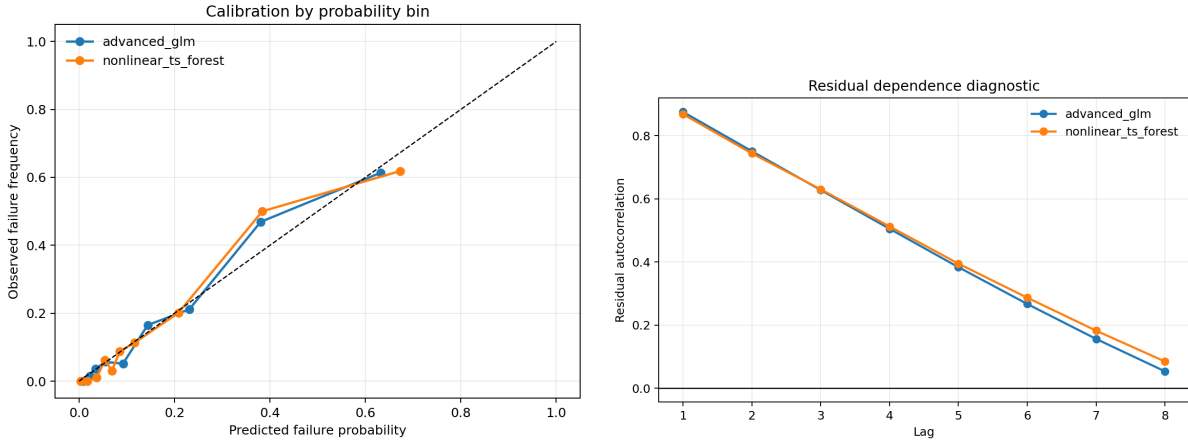


Figure 3: Calibration and residual dependence diagnostics for the strongest predictive models.

6 Maintenance Policy Comparison

Policies are evaluated in a continuing-operation environment in which each replacement starts a fresh machine. Preventive replacement costs 18 units, while failure-driven replacement costs 95 units.

Table 5: Average cost and event counts by maintenance policy.

policy	avg_reward	avg_cost	avg_replacements	avg_failures
risk_threshold_0.25	-77.6429	77.6429	2.7857	0.3571
q_learning	-100.4286	100.4286	4.3571	0.2857
age_threshold_75	-200.6429	200.6429	2.2857	2.0714
reactive	-203.5714	203.5714	2.1429	2.1429
dqn	-223.9286	223.9286	2.3571	2.3571

The tuned risk-threshold rule uses a threshold of 0.25. It achieves the lowest average cost in this experiment, which reinforces the main story of the project: once the risk estimate is informative, a simple decision rule can be highly effective.

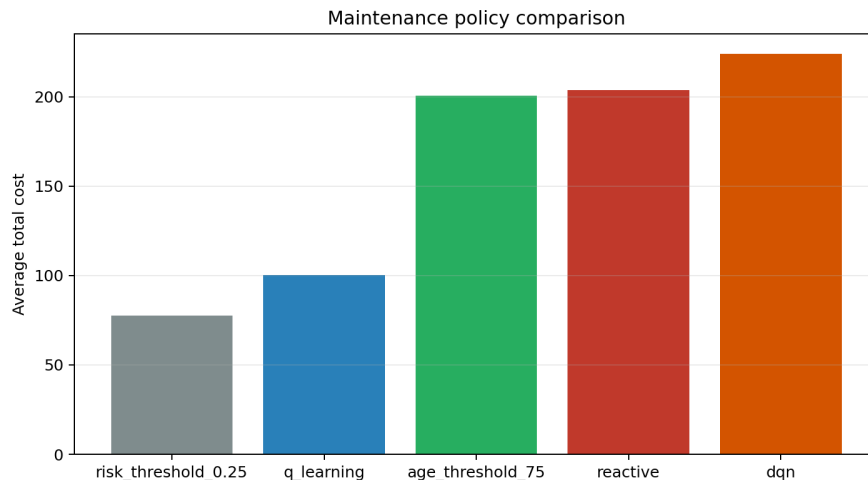


Figure 4: Average maintenance cost across competing policies.

7 Discussion

The strongest improvement in this project comes from better temporal representation rather than from a more complicated control algorithm. This is helpful for a course project because it keeps the main contribution easy to explain: rolling summaries, slopes, and lag structure turn noisy sensor streams into operationally useful risk estimates. The policy layer then demonstrates why that forecasting improvement matters.

8 Limitations and Future Work

The project uses synthetic data rather than a real industrial benchmark, and the RL layer is intentionally lightweight. Natural extensions include state-space or hidden Markov formulations, richer partially observed RL, and evaluation on datasets such as NASA C-MAPSS. The DQN baseline included in the repository is best treated as an extension rather than the main result.

9 Reproducibility

The full pipeline can be reproduced by installing the dependencies from `requirements.txt` and running `python run_project.py`. The script regenerates all tables, figures, and both report formats. The notebook in `notebooks/final_project_analysis.ipynb` provides a complete end-to-end analysis starting from raw synthetic data generation.